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# Evaluating the Economic Impacts of Light Rail by Measuring Home Appreciation: A First Look at New Jersey's River Line

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**Abstract**

Economic benefits are sometimes used to justify transport investments. Such was the case with the River Line of southern New Jersey, USA, which broke ground in 2000 and began operating in 2004. Recently, the line has been performing near full capacity and there is evidence that it has spurred development. Disaggregate data on owned-home appreciation are used to investigate the initial economic impacts of the line, looking carefully at non-linearity in the appreciation gradient, differential effects of station ridership and parking, redistribution of property appreciation gains and differences by property and neighbourhood type. At this time, the net impact of the line on the owned housing market is neutral to slightly negative. While lower-income census tracts and smaller houses seem to appreciate near the station, this may be a value transfer from farther-away properties not favoured with access. Few studies have previously looked for such effects.

## 1. Introduction

Transport investments are sometimes justified on the basis of their anticipated economic benefits. Such was the case for the River Line passenger rail service in southern New Jersey, USA. Opened in March 2004, the line stretches 34 miles between its termini in Camden and Trenton, two formerly industrial cities that were anchors of a thriving manufacturing economy in southern New Jersey many decades ago (Figure 1). There

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are 18 stops along the way, primarily in small towns near the Delaware River. The River Line has been called an 'interurban' railway because, while at points it acts like a light rail line with at-grade crossings at intersections, it reaches high speeds between town centres like a commuter rail line (Fazio, 2007). The line was financed by the state transport trust fund without federal match and with no contribution from local property or sales taxes.

Historically, the River Line corridor offered leisurely river town retreats for Philadelphia residents, with bucolic settings and recreational activities like sailing and swimming along the Delaware River (Albert and Albert, 2002). With the rise in automobile use, Philadelphians' preferences for recreational areas changed and the tourist economy of the river towns suffered. More recently, the river towns have attracted food processing and regional distribution centres to fill the economic void left after the decline of recreation, as the hinterlands of Philadelphia and New York still have many farms (NJDOT, 2008).

From its conception in the early 1990s, the River Line was controversial. A number of public officials from southern New Jersey hoped that it would help to revive the moribund river towns along state route 130 between Trenton and Camden, stimulating tourism from other parts of the state to some of the historical locations in those towns, and bringing visitors to an aquarium, an amphitheatre and minor league sports stadiums; and by capturing commuters to the government complex in Trenton, the Camden campus of Rutgers University and to points as far north as Manhattan via transfer

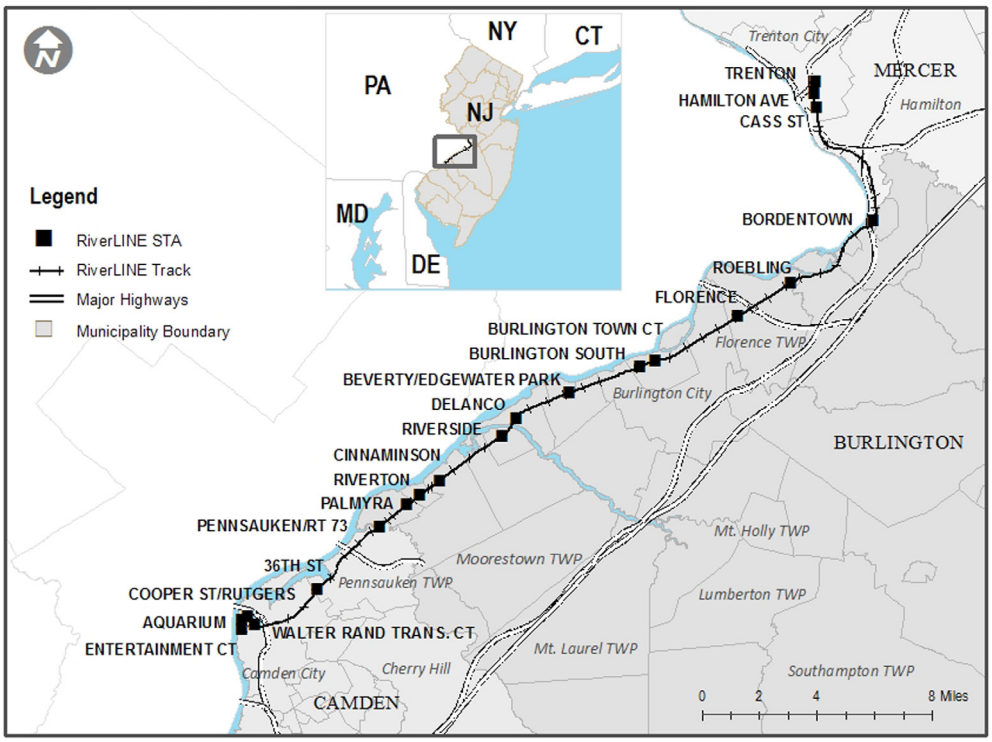



**Figure 1.** River Line and stations.

to the Northeast Corridor line. However, as a newspaper article noted, there were also concerns about negative impacts.

> Many 'not-in-my-backyard' fears have been expressed at NJTransit public hearings, including complaints that train service and stations will drive down property values, create unbearable noise and invite desperate criminals from downtrodden Camden to plunder the suburbs (Philadelphia Inquirer, 1995).

After arriving at a final alignment, the project was announced and groundbreaking commenced in May 2000. The line was eventually completed in March 2004 for a cost of $1.1 billion. Ridership was initially very low, at about 3000 trips per day. Within six months it had increased to 6000 trips per day, higher than the last pre-operation projections—although far below the first projections, for a different alignment, of more than 11 000 trips per day (Nigro, 2005). Usage has since increased to 9000 trips per day (Nussbaum, 2009), approaching the line's capacity, and significantly exceeding the expectations of critics of the line. The line currently operates more than 40 round trips per day.

Despite relatively low ridership in comparison with other rail lines in the US, anecdotal evidence and stories in the local press suggest that the River Line may be inducing new development and altering the economic makeup of the region. Hundreds of new condominium units have been proposed, permitted and, in some cases, constructed near River Line stations during this period. The increase in building permits between 2004 and 2007 was much higher for towns with stations than for towns elsewhere in the area, and there was an upturn in multifamily housing near the line.

At this early point in the River Line's history, observing the sales prices of nearby properties before and after the line's groundbreaking and opening, and comparing them via controlled analysis with comparable properties elsewhere, provides perhaps the best available measure of its economic impacts. In the remainder of the paper, we review relevant literature, describe our methods and data, and discuss the meaning of our results for other studies of this type and for policies about rail investments.

## 2. Literature Review

Transport investments have economic benefits primarily when they significantly improve access. Benefits tend to be highest near, but not too near, network access points such as rail stations or freeway ramps. Places very near the network may experience negative impacts such as noise, congestion, pollution and accidents (Kilpatrick et al., 2007). Changes in the value of properties near the transport improvements may reflect both benefits and costs.

Almost all property value studies of rail transit impacts have been conducted using a cross-sectional design. Here, we review studies of light rail systems, which are most relevant to our study. Many of these have tested for both the benefits and the costs associated with rail operation and some have used a rich set of control variables. A study of the Portland MAX light rail system found that the value of accessibility to the station generally exceeded the nuisance of the line (Chen *et al.*, 1998). The property premium compared with the low of the rent gradient was estimated at about 10.5 per cent. Another Portland study found a premium ranging between about 10 and 30 per cent for the western light rail corridor (Knaap *et al.*, 2001). A study of both light rail and commuter rail in 1999 in Santa Clara County found that land occupied by apartments near light rail stops was up to 45 per cent more valuable than comparison properties, while land near commuter rail had a premium of about 20 per cent (Cervero and Duncan, 2002a). The same authors found the

opposite for commercial properties: higher land values for commuter rail access than for light rail access (Cervero and Duncan, 2002b). The high premiums found by both studies probably reflect the focus on the parcel value rather than the whole property value, as well as the strong economic conditions and scarcity of housing in Santa Clara County at the time. Finally, a study of San Diego properties found significantly higher premiums for condominiums within a quarter-mile of light rail stations in San Diego than for single-family homes: 17 per cent versus 6 per cent (Duncan, 2008). This study is notable for distinguishing capitalisation effects for attached and detached owned housing.

East Coast studies include a study of MBTA's Fitchburg commuter line in Boston, which has similar operating characteristics to the River Line (Armstrong, 1995). Homes located in census tracts with rail stations had 6.7 per cent higher selling prices, but immediate proximity to the line (within 400 feet) coincided with a 20 per cent decrease in value, suggesting disamenity effects caused by frequent freight trains. A broader study in the Boston metropolitan area included seven communities, four with commuter rail service. It found a 10 per cent increase in value near stations and a penalty between $73 and $290 per 100 feet closer to the right-of-way (Armstrong and Rodriguez, 2006). A study of the Buffalo, NY, light rail line found a premium of between 2 and 5 per cent of value, with lower effects for properties in economically declining areas and higher effects in more prosperous areas (Hess and Almeida, 2007).

A cross-sectional study of Atlanta's rapid rail line MARTA is notable for its thoroughness, accounting for distance to stations, retail activity and crime. The positive effect of access to stations was generally greater than the negative effects of crime or the positive effects of retail, although within a quarter-mile radius some stations appeared to have net neutral or negative impacts (Bowes and Ihlanfeldt, 2001). Properties in higher-income neighbourhoods had larger capitalisation of proximity, consistent with Hess and Almeida (2007). Total effects varied considerably depending on census tract income, distance to downtown and distance to the station.

We turn now to describing three repeat-sales studies which are more robust to omitted variable problems and endogeneity. Gatzlaff and Smith (1993) estimated hedonic cross-sectional regressions, as well as the construction of a price appreciation index based on repeat sales, before and after the opening of the Miami Metrorail heavy rail rapid transit line. They found no strong evidence of effects of the line on single-family detached residential properties in either model, but their results are limited to a comparison of sales indexes. The remaining two studies are of disaggregate data for Chicago's Southwest Side Rapid Transit Line (SSRTL), an elevated rapid rail line that reduced transit travel times between Midway Airport and downtown by a third and carried 28 000 riders per day. McDonald and Osuji (1995) found an increase of 17 per cent in value for properties within a half-mile of stations by examining comparative parcel sales from 1980 to 1990, prior to the line's 1993 opening, using a dataset of estimated values for 79 residential blocks. McMillen and McDonald (2004) studied the same line cross-sectionally and longitudinally, using a larger dataset of 4000 properties within 1.6 miles of the line. They found a varying amount of capitalisation depending on the year of second sale, ranging from 4 per cent 10 years before the line opened to 19 per cent for a five-year period before and after the line's opening, and averaging about 10 per cent after the opening of the line.

## 3. Methods and Data

### 3.1 Methods

We analysed property appreciation rates as a function of rail access along with a large set of control variables. Repeat-sales data on

properties are potentially very good indicators of economic impacts. In rare instances where repeat sales data have been used, they have typically been either with very small datasets (for example, McDonald and Osuji, 1995) or to construct price indexes (for example, Gatzlaff and Smith, 1993; Wang and Zorn, 1997; McMillen, 2003). The study with methods most comparable to this one is McMillen and McDonald (2004).

Almost all other transport-related property value studies have examined variation between properties rather than over time. Unobserved or unobservable variables can potentially bias coefficient estimates in such models. Endogeneity problems are also reduced with repeat sales methods. Cross-sectional studies can only indirectly control for the fact that rail systems and routes may be routed through places that are more economically depressed (for example, right of way is cheaper there) or along existing rights of way because of demand (implying positive economic activity). In the absence of a natural experiment, repeat sales are a good way to control for endogeneity and omitted variable bias.

There are some potential disadvantages to using repeat sales as well. First, there can be selection bias if properties that sell more frequently are also systematically and unobservably different from properties that sell less frequently. Evidence on this is mixed. While some have found that homes with more sales appreciate faster (for example, Case *et al.*, 1997), others suggest that we know little about how unobserved properties differ from observed ones (Wang and Zorn, 1997). Secondly, confining estimation to properties with more than one sale substantially reduces sample size, making estimates of impact less precise (Case and Quigley, 1991). Finally, properties may change significantly between transactions (Goetzmann and Spiegel, 1995); they may be improved, structurally modified or degrade due to poor maintenance. However, this only creates a potentially significant bias in our estimates if properties near the River Line are more or less likely than properties elsewhere to have been significantly improved or better maintained, and if that likelihood is uncorrelated with the numerous variables used in the models. We know of no reason to suspect this is the case.

Unlike any previous repeat-sales study that we are aware of, we include property characteristics in our model to control for the possibility that there are differential appreciation rates by property type and that those property types are distributed non-randomly with respect to rail access. This also controls for sample selection from repeat sales, to the extent that sample selection is over observable variables.

Our dependent variable is the logged ratio of before and after sales prices (consistent with Goetzmann and Spiegel, 1995; McDonald and Osuji, 1995; and McMillen and McDonald, 2004)

$$\log\left(\frac{P_2}{P_1}\right) = \alpha + \beta X + \gamma N + \delta S + \lambda C + \eta_1 + \eta_2 + \nu + \mu + \varepsilon$$

where, $P_1$ is the observed property price at time 1 (before groundbreaking or operation); and $P_2$ the observed property price at time 2 (after groundbreaking or operation). The logged ratio is equivalent to the percentage change in the property value between time 1 and time 2. Independent variables are: $X$, a vector of spatial variables measuring access to River Line stations; $N$, a vector of spatial variables representing possible nuisance effects of the line; $S$, a vector of accessibility variables such as bus stop distance, freeway access, freeway right-of-way proximity, other rail system access and distance to major job centres; $C$, a vector of control variables for properties, including property characteristics and neighbourhood characteristics; $\mu$, a fixed

municipality effect; $\nu$, a fixed county effect; and $\eta_1$ and $\eta_2$, fixed effects of the years of first sale and of last sale.

Using the logged ratio of sales prices at two times, before and after the line's announcement or opening, is preferable to using yearly property appreciation because we expect a one-time capitalisation effect (Goetzmann and Spiegel, 1995). We account for yearly differences and yearly appreciation by including variables for duration of time on the market, year of first sale and year of last sale.

Few studies have included distance threshold effects or non-linearity in considering how station proximity affects property prices. People may value walking distance or short driving distance access much more than stations that are farther away, and they may be more likely to view stations very near them as nuisances because of traffic and noise. A notable early study found that a reciprocal function of distance was helpful in accounting for negative externalities and the positive accessibility value of stations (Skaburskis, 1982). We investigated such effects using distance threshold measures before and after groundbreaking and operation.

Transport improvements may result not just in land value increases for properties whose accessibility has improved, but also a reduction in property values for places that do not benefit from improved access, as their relative importance declines (Mohring, 1961). To discover any evidence of value transfer, we compared distance threshold coefficients for nearby properties in comparison with controls farther away from stations.

There is also some evidence from cross-sectional studies that capitalisation effects may vary among population sub-groups or property types (Adair *et al.*, 2000; Baum-Snow and Kahn, 2000; Weinberger, 2001; Duncan, 2008), suggesting the need for a focus on different property sub-markets. Rental housing and non-residential office buildings might be more likely to benefit from rail access, but data on such properties were not available for the study area. We instead carried out some tests of sub-markets within the owned housing market that might be thought to respond more readily to transit access: smaller houses and homes in lower-income census tracts.

### 3.2 Data Assembly

Sales values and property characteristics at sale are from two sources for different periods of time, matched using parcel identifiers and geocodes into a single dataset covering all properties sold in a four-county region. Three of the counties contain River Line stations; the exception, Gloucester County, is included in order to distinguish River Line access from views of and access to the Delaware River. For 1989 to 2000, the data are from the state of New Jersey via a private vendor, Econsult, Inc. For 2001 to 2007, they come from purchased realtor listings via TREND Multiple Listing Service of Philadelphia. The possibility of systematic differences between the data sources is controlled with a set of dummy variables representing the years of first and second sale. Tests on an overlapping set of data found that in 99.4 per cent of cases sales prices were identical and dates were either identical or within two months of each other.

The NJ/Econsult dataset contains basic property sales data from county ledgers, including selling prices for each property for up to seven transactions. The data include a very small number of property characteristics and address information, but the quality is poor and many records are missing values. This dataset initially had 352 106 records. A number of sales were carried out below market value for various reasons, such as non-arms-length transactions between family members. We selected a cut-off point of $10 000 as a reasonable estimate for a market transaction in this database, removing 106 259 properties, some of which were apparently vacant land classified as residential. After

discarding about 50 000 additional unusable records with significant missing values, we were left with 191 086 properties in the four-county region, most with multiple transactions per property.

The second dataset from the Multiple Listing Service contains brokered residential property transactions, including sales prices as well as the number of bedrooms, number of bathrooms, lot size, age, parcel information and address, for a total of 153 360 records, of which 18 010 were duplicate transactions. After converting the dataset to a multiple-sales-per-property format, we were left with 115 872 properties. We merged this dataset with the NJ/Econsult dataset and deleted properties that did not sell either before and after groundbreaking in 2000, or before and after operation in 2004. This left 48 968 properties, about 40 per cent of the total. Discarding duplicate transaction records left us with 40 161 properties with complete attributes from both datasets. Geocoding was successful for 95.3 per cent of these properties, leaving 38 050. We excluded properties which sold inside a window three months before and after groundbreaking or operation, and were left with 32 384 (85 per cent) of the geocoded properties. Finally, after discounting all values for inflation, we removed 914 properties (2.8 per cent of otherwise eligible observations) with sales values of less than $10 000 or more than $2 million in either year, or that lost more than 50 per cent of their value or gained more than three times their value. The truncation of the dataset was to remove outliers that may not have been subject to arms-length transactions. Our final estimation dataset size was 31 470 properties.

### 3.3 Variable Construction

The variables of most interest to this study are distances along the road network ('network' distances) from properties to River Line stations, to test for the accessibility value of the line; and straight-line ('airline') distances to its right-of-way and to its whistle-blowing points, to test for any nuisance impacts of the line (Tables 1 and 2). These measures were constructed using GIS software with the geocoded parcels and a network analysis routine. Network distances are the best estimate of actual walking or driving

**Table 1.** Variable descriptions

| *Variable* | *Description* |
|---|---|
| *Dependent variable* | |
| log_valratio | Log of ratio of second and first sales prices |
| *Independent variables of interest* | |
| dist2sta, dist2sta_**gb**, dist2sta_**op** | Network distance to the nearest River Line station |
| dsq2sta_net, dsq2sta_net_**gb**, dsq2sta_net_**op** | Squared network distance to nearest River Line station |
| RLqmi_**gb**, RL1mi_**gb**_rg , ... , RL5mi_**gb**_rg ; RLqmi_**op**, RL1mi_**op**_rg , ..., RL5mi_**op**_rg | Indicator variables for network distance rings to 1/4, 1/2, 1, 2, 3, 4 and 5 miles from a River Line station |
| WHemi_**gb**...WH1mi_**gb**_rg, WHemi_**op**... WH1mi_**op**_rg | Indicator variables for airline distance rings to 1/8, 1/4, 1/2 and 1 mile from a River Line whistle-blowing point |
| TRemi_**gb**...TR1mi_**gb**_rg ; TRemi_**op**... TR1mi_**op**_rg | Indicator variables for airline distance rings to 1/8, 1/4, 1/2 and 1 mile from River Line track right-of-way |
| parkstn_qmi...parkstn_1mi_rg, boardings_qmi... boardings_1mi_rg | Parking spaces and daily boardings at nearest station within a 1/4, 1/2 or 1 mile ring (post-operation only) |

**Table 1.** *(Continued)*

| *Variable* | *Description* |
|---|---|
| *Spatial control variables* | |
| ORemi, ORqmi_rg, ORhmi_rg, OR1mi_rg | Dummy airline distance thresholds of 1/8, 1/4, 1/2 and 1 mile from Atlantic Ocean, Delaware River or Delaware Bay |
| dist2exit | Network distance to freeway or highway entrance/exit |
| dist2hwy, dist2fwy | Airline distance to nearest highway/freeway right-of-way |
| dist2bstop | Network distance to nearest bus stop |
| dist2phi, dist2nyc, dist2trent, dist2camd | Highway network distance to Philadelphia, Manhattan, Trenton and Camden CBDs |
| dist2NEC, dist2PATCO, dist2ACL | Network distance to Northeast Corridor, Port Authority Transit Corporation or Atlantic City Line rail station |
| dist2river | Airline distance to coastline (generalised spatial trend) |
| X, Y | Latitude, longitude (generalised spatial trend) |
| *Property and neighbourhood characteristics* | |
| bedrooms, fullbaths, halfbaths, totrooms, lotsize, design_2...11, AC, age, age2 | Characteristics of home at second sale (AC = air conditioning; age2 = squared age of home; design_2…11 = set of design variable dummies—for example, 2-storey) |
| medhhinc, rentsh, vacsh, afamsh | Median household income, share of housing that was vacant, share of housing that was renter-occupied, percentage of African Americans in the property's 2000 census tract or block group |
| TP0506 | Average district elementary test score for 2005/06 school year |
| SAT | Average SAT score of high school district (2005/06) |
| colleg4 | Share of HS graduates attending 4-year college (2005/06) |
| *Time controls and fixed effects* | |
| elapsed | Years elapsed since the most recent sale |
| y1990_first...y1999_first; y2002_second...y2007_second | Dummies for fixed appreciation effects of varying market conditions from year to year (year of first and second sale) |
| ppgb, ppop | Dummy variables indicating if the property sold before/after groundbreaking or before/after operation of the River Line |
| Mercer, Camden, Gloucester, Burlington | Fixed county effects (3 dummies, Burlington omitted) |
| Princeton, Trenton, Burlington_City, ... , … , … | Fixed municipality effects (111 dummies) |

*Notes*: 'missing value' dummies included for totrooms, lot size, AC, age, medhhinc, afamsh, TP0506, SAT and colleg4. '**_op**' and '**_gb**' affixes denote variables with non-zero values for properties that sold either before and after **g**round**b**reaking or before and after the line started **op**erating (see text).

distance to access stations from properties and can vary significantly from airline distance. We also measured network distances to highway and freeway exits, airline distance to highway and freeway right-of-way (to control for highway traffic impacts), network distances to bus stops, network distances to stations on other rail systems and freeway network distances to

**Table 2.** Variable summary statistics, part 1

| *Variable* | *Mean* | *SD* | *Minimum* | *Maximum* |
|---|---|---|---|---|
| Log of ratio of second and first sales prices | 0.36 | 0.28 | –0.69 | 1.39 |
| Network distance to nearest River Line station (miles) | 9.18 | 5.77 | 0.00 | 46.14 |
| Network distance to nearest River Line station properties sold pre-groundbreaking (miles) | 9.21 | 5.77 | 0.02 | 46.14 |
| Network distance to nearest RiverLine station properties sold post-operation (miles) | 9.19 | 5.77 | 0.005 | 46.14 |
| Network distance to highway entrance/exit | 16.83 | 4.84 | 0.44 | 20.00 |
| Airline distance to highway entrance/exit | 2.38 | 2.63 | 0.00 | 19.40 |
| Airline distance to freeway entrance/exit | 13.80 | 6.59 | 0.00 | 20.00 |
| Network distance to nearest bus stop | 7.33 | 6.68 | 0.00 | 20.00 |
| Highway network distance to Philadelphia CBD | 229.87 | 103.25 | 30.41 | 812.85 |
| Highway network distance to Manhattan CBD | 873.63 | 150.73 | 512.93 | 1,226.63 |
| Highway network distance to Trenton CBD | 274.38 | 122.23 | 2.31 | 543.27 |
| Highway network distance to Camden CBD | 171.93 | 106.45 | 2.04 | 496.88 |
| Network distance to Northeast Corridor rail station | 19.75 | 1.50 | 1.05 | 20.00 |
| Network distance to PATCO rail station | 19.16 | 2.92 | 0.02 | 20.00 |
| Network distance to Atlantic City Line rail station | 19.76 | 1.36 | 2.55 | 20.00 |
| Airline distance to river coastline | 17.52 | 5.15 | 0.07 | 20.00 |
| X-co-ordinate (metres in 10 000s) | 37.43 | 4.72 | 24.65 | 52.11 |
| Y-co-ordinate (metres in 10 000s) | 40.55 | 6.34 | 25.39 | 57.66 |
| Boardings at nearest station within 1/4 mile (100s) | 2.72 | 1.08 | 0.82 | 4.80 |
| Parking at nearest station within 1/4 mile (100s) | 1.24 | 1.25 | 0.00 | 3.67 |
| Boardings at nearest station 1/4–1/2 mile (100s) | 2.99 | 1.93 | 0.33 | 14.18 |
| Parking at nearest station 1/4–1/2 mile (100s) | 1.28 | 1.26 | 0.00 | 3.67 |
| Boardings at nearest station 1/2–1 mile (100s) | 3.08 | 2.09 | 0.33 | 14.53 |
| Parking at nearest station 1/2–1 mile (100s) | 1.62 | 1.57 | 0.00 | 5.89 |

N*otes*: All distances in 0.1 mile units unless otherwise specified. Summary statistics for boardings and parking variables and properties sold pre-groundbreaking and post-operation are for relevant observations only.

the main job centres in the region. To control for the demand for waterfront property, we included the aerial straight-line distance to the Delaware River and tributaries as well as to the Atlantic Ocean and Delaware Bay. Finally, we attached NJ Transit data on the number of parking spaces at the nearest stations in 2006 and on the number of boardings at each station in 2004, 2005, 2006 and 2007, corresponding to the year of second sale.

Properties were categorised as selling either: [a] first before groundbreaking and then after commencement of operation (43 per cent of properties); [b] first between groundbreaking and operation and again after operation (28 per cent of properties); or [c] first before groundbreaking and again between groundbreaking and operation (29 per cent of properties). Since properties may fall in more than one category but only one sales pair is used for each property, we categorised properties first as category [a], then as category [b] and otherwise as category [c]. There are two kinds of variables representing distances to stations, track and whistle-blowing points: those that are measured for properties sold before and after groundbreaking in May 2000; and, those that are measured for property sold before and after operation commenced in March 2004 (see Table 1).[1] For category [a] properties, both types of variable are non-zero.

Base model control variables (included in all regressions but not shown) include standard housing unit characteristics at the time of

the most recent sale; years elapsed since the most recent sale; years elapsed since 2000, the year of River Line groundbreaking; dummies representing the years of both the first and second sale, to account for fixed appreciation effects associated with varying market conditions from year to year; latitude and longitude, to account for any north–south or east–west spatial trends in prices not captured by other spatial variables; and dummies for four counties and 111 municipalities to account for possible fixed effects due to differences in zoning regime, blight, local market conditions and other community characteristics (Tables 1 and 3). In addition, data from the 2000 Census on the tract's median income, the share of the population that was African American, the share of housing that was rented and the share of housing that was vacant were spatially joined to the geocoded properties using Census shapefiles. Finally, elementary and high school test score data were attached via a spatial join of the properties to two different school district shapefiles.

Of the sample, 1922 units (6.1 per cent) are within a one-mile network distance of a River Line station, in 16 municipalities; 740 units (2.4 per cent) within a half-mile, in 13 municipalities; and 146 (0.5 per cent) within a quarter-mile, in 12 municipalities.

## 4. Analysis

In the 'full models' (Table 4) we used all properties in the dataset and in subsequent models (Table 5) we investigated subsets split by Census tract median income and housing unit size. The base model (not shown) includes 59 independent variables along with over 100 municipality dummy variables, including property characteristics, Census tract and school district characteristics, and fixed effects for municipality, county and year of first and second sale, as described earlier. We ran ordinary least squares regressions using a correction for heteroscedasticity to account for a varying number of years between sales. Alternative model forms were tested and

**Table 3.** Variable summary statistics, part 2

| *Variable* | *Mean* | *SD* | *Minimum* | *Maximum* |
|---|---|---|---|---|
| Number of bedrooms | 3.1 | 0.83 | 1 | 11 |
| Number of full bathrooms | 1.6 | 0.61 | 0 | 8 |
| Number of half bathrooms | 0.54 | 0.53 | 0 | 5 |
| Total rooms | 7.2 | 1.80 | 3 | 15 |
| Lot size (1000s of square feet) | 10.0 | 15 | 0.10 | 752 |
| Air conditioning (yes = 1) | 0.77 | 0.42 | 0.00 | 1 |
| Age of building (decades) | 3.7 | 2.7 | 0 | 35 |
| Age of building squared (centuries) | 21.0 | 36 | 0 | 1,225 |
| Median household income ($1000s) | 61.4 | 22.5 | 2 | 200 |
| Rental rate | 0.19 | 0.17 | 0 | 1 |
| Vacancy rate | 0.04 | 0.04 | 0 | 0.59 |
| African American population share | 0.15 | 0.18 | 0.001 | 0.96 |
| Elementary school test scores | 60.3 | 5.9 | 43.0 | 77.8 |
| SAT scores | 1470 | 157 | 1136 | 1774 |
| College graduates (percentage) | 52.9 | 12.9 | 19.8 | 91 |
| Number of years between first and last sale | 6.7 | 3.1 | 0.75 | 16 |
| First sale before groundbreaking (yes = 1) | 0.28 | 0.45 | | |
| Second sale after operation began (yes = 1) | 0.29 | 0.45 | | |
| Property in Mercer County | 0.17 | 0.37 | | |
| Property in Camden County | 0.34 | 0.48 | | |
| Property in Gloucester County | 0.15 | 0.36 | | |

the log-linear form performed better than alternative forms such as log-log.

### 4.1 Full Model Results

We began with a model to test whether network distance to stations after groundbreaking in 2000 and after opening in 2004 is associated with property appreciation (Table 4, model 1). We included four distance variables to distinguish the effects of station access to River Line stations after groundbreaking from the effects of station access after line operation commenced, including both a linear and squared term for distance.[2] We found a statistically significant and positive coefficient on distance before groundbreaking, of about 0.1 per cent per tenth of a mile, implying a negative effect of River Line stations on property values prior to opening in 2004. We found a larger negative coefficient on post-operation distance, of about 0.2 per cent per tenth of a mile distance from a rail station, implying that access to stations in the post-operation period is associated with property appreciation—more in accordance with expectation. By adding the coefficients for the two distance variables together and testing for statistical significance (results not shown in Table 4), it is possible to infer the net effect of both groundbreaking and operation: one-tenth of a per cent property appreciation per tenth of a mile nearer the station (statistically significant at the 99 per cent confidence).

In model 2, we added the distance threshold variables to see whether the relationships found between station access and property appreciation were non-linear or restricted to shorter distances. We expected that property value increases might be felt primarily within walking distance of rail, as would any nuisance effects; and that the bid–rent gradient might be flatter beyond that distance. We also sought to measure any redistribution of gains between nearer and farther away properties. We also tested spline variables allowing for different bid–rent gradient slopes between each of those thresholds, as well as a negative reciprocal, but these variants were not statistically superior to this simpler specification.

The results of model 2 are consistent with model 1 but more highly statistically significant, again showing large differences between groundbreaking and announcement within five miles of stations and, more importantly, picking up larger negative impacts and high non-linearity within five miles of stations. Properties selling after groundbreaking depreciated between about 15 and 8 per cent depending on distance from the station out to about three miles away, with a smaller negative effect of 2.4 per cent in the four-to-five mile band. Meanwhile, properties that sold first after groundbreaking and again after operation had a relative increase in value ranging from 12 to 14 per cent within a half-mile of stations, to about 11 per cent between a half-mile and a mile, to between 4 and 5 per cent for properties in the one-to-four mile bands, with no statistically significant effect in the four-to-five mile band. Note also that in this model, the coefficients on continuous distance to the station both before and after operation are negative, relatively small and statistically indistinguishable from each other, although statistically significant.[3]

Also in model 2, we tested whether there is any net correlation of property appreciation with the River Line station threshold variables by adding post-groundbreaking and post-operation coefficients for the variables and testing for statistical significance. For properties right near the station out to one mile away, there is no net effect: the sums are small and statistically indistinguishable from zero. For properties between one and four miles away, the net effect is negative: between -3.2 and -5.2 per cent (99 per cent confidence).

Could these apparent net neutral or negative effects of the River Line be affected by local nuisances due to ridership and parking—crime, foot traffic and automobile traffic? We added variables for the number of parking places and

**Table 4.** Home value appreciation models, all properties ($N$ = 31 470)

| | *Model 1 Distance to stations, before and after operation* | *Model 2 Model 1 + distance thresholds* | *Model 3 Model 2 + parking and boardings* | *Model 4 Model 3 + distance to track, whistle points, river* | *Model 5 Model 4 + other access measures* |
|---|---|---|---|---|---|
| *Distance to station* | | | | | |
| First sale pre-groundbreaking (mile/10) | **0.001** | −0.001 | −0.001 | −0.001 | −0.001 |
| Second sale post-operation (mile/10) | **−0.002** | −0.001 | −0.001 | −0.001 | ns |
| Squared, first sale pre-groundbreaking (1000 miles$^2$) | **−0.060** | ns | ns | ns | ns |
| Squared, second sale post-operation (1000 miles$^2$) | **0.080** | **0.035** | **0.034** | **0.034** | 0.027 |
| *Station within ...* | | | | | |
| 1/4 road mile, post-groundbreaking | | **−0.147** | **−0.141** | −0.13 | −0.123 |
| 1/4 road mile, post-operation | | **0.119** | 0.229 | **0.279** | **0.291** |
| 1/4–1/2 road mile, post-groundbreaking | | **−0.139** | **−0.139** | **−0.165** | **−0.159** |
| 1/4–1/2 road mile, post-operation | | **0.136** | 0.0827 | 0.107 | 0.113 |
| 1/2–1 road mile, post-groundbreaking | | **−0.126** | **−0.127** | **−0.16** | **−0.154** |
| 1/2–1 road mile, post-operation | | **0.106** | ns | ns | ns |
| 1–2 road miles, post-groundbreaking | | **−0.105** | **−0.104** | **−0.12** | **−0.116** |
| 1–2 road miles, post-operation | | **0.053** | **0.055** | **0.057** | **0.062** |
| 2–3 road miles, post-groundbreaking | | **−0.084** | **−0.084** | **−0.082** | **−0.079** |
| 2–3 road miles, post-operation | | **0.038** | **0.039** | **0.039** | **0.042** |
| 3–4 road miles, post-groundbreaking | | **−0.086** | **−0.085** | **−0.083** | **−0.082** |
| 3–4 road miles, post-operation | | **0.054** | **0.054** | **0.055** | **0.057** |
| 4–5 road miles, post-groundbreaking | | −0.024 | −0.023 | −0.023 | −0.024 |
| 4–5 road miles, post-operation | | ns | ns | ns | ns |
| *Boardings at station within ...* | | | | | |
| 1/4 mile, year of second sale | | | −0.053 | −0.062 | −0.064 |
| 1/4 to 1/2 mile, year of second sale | | | ns | ns | ns |
| 1/2 to 1 mile, year of second sale | | | **0.017** | **0.018** | **0.018** |
| *Parking spaces at station within ...* | | | | | |
| 1/4 mile | | | ns | ns | ns |
| 1/4–1/2 mile | | | ns | ns | ns |
| 1/2–1 mile | | | ns | ns | ns |
| *Whistle point within ...* | | | | | |
| 1/8 mile, post-groundbreaking | | | | ns | ns |
| 1/8 mile, post- operation | | | | ns | ns |

**Table 4.** *(Continued)*

| | *Model 1 Distance to stations, before and after operation* | *Model 2 Model 1 + distance thresholds* | *Model 3 Model 2 + parking and boardings* | *Model 4 Model 3 + distance to track, whistle points, river* | *Model 5 Model 4 + other access measures* |
|---|---|---|---|---|---|
| 1/8–1/4 mile, post-groundbreaking | | | | ns | ns |
| 1/8–1/4 mile, post- operation | | | | ns | ns |
| *Track within ...* | | | | | |
| 1/8 mile post-groundbreaking | | | | ns | ns |
| 1/8 mile, post- operation | | | | ns | ns |
| 1/8–1/4 mile, post-groundbreaking | | | | ns | ns |
| 1/8–1/4 mile, post- operation | | | | ns | ns |
| *River/ocean within ...* | | | | | |
| 1/8–1/4 mile | | | | **0.039** | 0.035 |
| 1/2–1 mile | | | | ns | –0.016 |
| *Network distance to ...* | | | | | |
| Major highway exits | | | | | ns |
| Nearest ACL station | | | | | **–0.001** |
| Nearest bus stop | | | | | ns |
| Constant | **–1.179** | **–1.077** | **–1.060** | **–1.020** | **–1.542** |
| $R^2$ | 0.383 | 0.386 | 0.386 | 0.388 | 0.388 |

*Notes*: Variables significant at $p < 0.01$ in **bold**; variables significant at $p < 0.05$ or $p < 0.1$ in regular type; ns = not significant at 0.01 level, suppressed. Distances, unless otherwise noted, in tenths of a mile. Insignificant and not shown: airline distance over 1/4 mile to track and whistle points; river/ocean within 1/8 mile and 1/4–1/2 mile; network distances to PATCO station, NEC station, Philadelphia, Trenton, Camden and New York City CBDs, airline distance to nearest highway or freeway. Control variables not shown: all variables in Table 3; 105+ municipality dummy variables.

number of passenger boardings for nearby stations within a quarter-mile, half-mile and a mile (model 3). While the parking variables were not statistically significant, two of the boardings variables were. Within a quarter-mile, stations with more riders have lower property appreciation. The coefficients on the distance threshold variables for post-operation sales change significantly, becoming larger and more positive, although somewhat less statistically significant. There are 104 properties within a quarter-mile of stations that sold after Line operation began and stations near those properties ranged from 82 to 480 boardings per day. Ignoring boardings, the net correlation of property appreciation with River Line station proximity for properties within a quarter-mile is the negative 14 per cent post-groundbreaking plus the positive 23 per cent increase post-operation, or a 9 per cent property value increase. However, since the mean number of boardings at stations within a quarter-mile is 272, the average disbenefit associated with boardings is 2.72 × 5.3 per cent, or 14.4 per cent. Therefore the net value impact for the average property sold prior to groundbreaking and then after opening is about –5 per cent (9 minus 14.4 per cent).

In the same model, between a half-mile and a mile away, boardings are positively correlated with property appreciation. This could be because high-ridership stations have

on average more accessibility value and some have better amenities such as concurrent streetscape improvements and landscaping. These characteristics might be valuable to properties that are not close enough to suffer the localised nuisance effects of high ridership.

Are these models misspecified because of track noise or whistleblowing impacts of River Line vehicles? In the next regression (model 4) we added several additional spatial variables: distance to whistle points, distance to River Line track and distance to the Delaware River. These variables are meant to distinguish the possible negative effects of whistles and operation noise and the possible effects of river access (including appreciation due to views, or depreciation due to flooding) and views from any measured effects of access to River Line stations.[4] We included measures both before and after operation commenced, to account for the possibility that home buyers might anticipate such impacts. We did not find negative impacts of whistleblowing or of track proximity. We do find that some of the measured positive effects of River Line station access, post-operation, become larger when the other characteristics are controlled. The other post-operation River Line accessibility coefficients are not much affected. The coefficients for river or ocean within one-eighth and one-quarter mile are positive and significant, suggesting that the amenity value of river access exceeds flooding risk at this distance.

Finally, in model 5, we add a number of additional accessibility measures: distance to other rail stations, to major central business districts (Manhattan, Philadelphia, Camden and Trenton) and to highway entrances, transit stops and highway right-of-way. Including these variables only marginally changes coefficients on the main variables of interest and they are for the most part insignificant and are not shown in Table 4 to save space.

### 4.2 Subset Model Results

We compared the relative effects of the River Line for homes in lower- and higher-income areas and for homes of different sizes to distinguish between different kinds of property that might be thought more or less likely to benefit from rail access. First, we ran regressions separately for lower-income and higher-income Census tracts (Table 5, columns 1 and 2). Households of lower income are more likely to use transit than to drive, all else equal. It is also possible that the River Line's operation has an amenity effect for poorer neighbourhoods but is not seen as an improvement by higher-income neighbourhoods. The median household income of Census tracts represented in the four-county area was $56 833 in the 2000 Census and the sample is split roughly in two around that point.

About 90 per cent of properties within a mile of stations are in low-income Census tracts. For these properties (Table 5, column 1), within a quarter-mile of stations there is a large and significant positive property appreciation estimate of *net* positive 35 per cent (40—5.04). However, for properties farther than one-quarter mile away the *net* estimate is neutral and, in the two to three mile radius, the estimate is negative. This suggests that the line may have redistributed property appreciation gains from properties farther away from stations to properties near stations (consistent with Mohring, 1961). Note that station boardings for properties within a quarter-mile are associated with a significant reduction in property value, but the net effect remains positive and statistically significant for almost all houses.

For the small number of houses in higher-income Census tracts (Table 5, column 2), having a River Line rail station within a quarter-mile is associated with a net negative but statistically insignificant 58 per cent reduction in property value. Boardings are associated with slightly higher appreciation for this group of houses but in no cases does the net effect exceed zero. Station parking for properties between one-quarter and one-half

**Table 5.** Home value appreciation models, by home type (varying sample sizes)

| | *Lower-income tracts (n = 15 469)* | *Higher-income tracts (n = 16 001)* | *1-2 bedroom houses (n = 6 425)* | *3-bedroom houses (n = 15 900)* | *4+-bedroom houses (n = 9 140)* |
|---|---|---|---|---|---|
| *Distance to station* | | | | | |
| First sale pre-groundbreaking | ns | ns | ns | ns | ns |
| Second sale post-operation | ns | **–0.002** | ns | ns | –0.001 |
| Squared (1000 miles$^2$), first sale pre-groundbreaking | ns | ns | ns | ns | ns |
| Squared (1000 miles$^2$), second sale post-operation | ns | **0.065** | ns | ns | **0.065** |
| *Station within ...* | | | | | |
| 1/4 road mile, post-groundbreaking | ns | ns | ns | –0.150 | ns |
| 1/4 road mile, post-operation | **0.400** | –0.42 [ns] | ns | **0.410** | ns |
| 1/4–1/2 road mile, post-groundbreaking | ns | –0.16 | ns | –0.150 | –0.170 |
| 1/4–1/2 road mile, post-operation | ns | ns | ns | 0.250 | ns |
| 1/2–1 road mile, post-groundbreaking | ns | **–0.180** | ns | **–0.150** | –0.160 |
| 1/2–1 road mile, post-operation | ns | ns | ns | 0.12 | ns |
| 1–2 road miles, post-groundbreaking | ns | **–0.099** | ns | **–0.099** | **–0.120** |
| 1–2 road miles, post-operation | ns | ns | ns | **0.094** | ns |
| 2–3 road miles, post-groundbreaking | –0.047 | –0.049 | ns | ns | **–0.120** |
| 2–3 road miles, post-operation | ns | ns | ns | ns | **0.087** |
| 3–4 road miles, post-groundbreaking | ns | **–0.068** | –0.070 | **–0.061** | **–0.094** |
| 3–4 road miles, post-operation | ns | 0.034 | ns | **0.058** | **0.080** |
| 4–5 road miles, post-groundbreaking | 0.038 | **–0.041** | ns | ns | –0.036 |
| 4–5 road miles, post-operation | –0.033 | ns | ns | ns | ns |
| *Boardings at station within ...* | | | | | |
| 1/4 mile | –0.084 | ns | ns | –0.08 | ns |
| 1/2–1 mile | 0.015 | 0.065 | 0.018 | ns | ns |
| *Parking spaces at station within ...* 1/4–1/2 mile | ns | –0.081 | ns | ns | ns |
| *Whistle point within ...* | | | | | |
| 1/8 mi, post-groundbreaking | ns | ns | ns | ns | ns |
| 1/8 mi, post-operation | ns | ns | ns | ns | ns |

*(Continued)*

**Table 5.** *(Continued)*

| | *Lower-income tracts (n = 15 469)* | *Higher-income tracts (n = 16 001)* | *1-2 bedroom houses (n = 6 425)* | *3-bedroom houses (n = 15 900)* | *4+-bedroom houses (n = 9 140)* |
|---|---|---|---|---|---|
| 1/8–1/4 mile, post-groundbreaking | ns | 0.260 | ns | ns | ns |
| 1/8–1/4 mile, post-operation | ns | ns | ns | ns | ns |
| 1/4–1/2 mile, post-groundbreaking | ns | 0.200 | ns | ns | –0.250 |
| 1/4–1/2 mile, post-operation | ns | –0.220 | ns | ns | ns |
| *Track within ...* | | | | | |
| 1/8 mile, post-groundbreaking | ns | ns | ns | ns | ns |
| 1/8 mile, post-operation | ns | ns | ns | ns | ns |
| 1/4–1/2 mile, post-groundbreaking | ns | ns | ns | ns | ns |
| 1/4–1/2 mile, post-operation | ns | 0.190 | ns | ns | ns |
| *River/ocean within ...* | | | | | |
| 1/8 mile | ns | –0.052 | ns | ns | ns |
| 1/8–1/4 mile | 0.049 | ns | 0.084 | ns | 0.062 |
| 1/4–1/2 mile | ns | ns | ns | ns | 0.058 |
| Airline distance to nearest state highway | ns | ns | 0.004 | ns | ns |
| Airline distance to freeway | ns | 0.001 | –0.003 | ns | 0.001 |
| *Network distance to major highway exits ...* | ns | ns | 0.003 | ns | –0.001 |
| Nearest ACL station | –0.001 | ns | **–0.002** | 0.00 | **0.001** |
| Nearest PATCO station | 0.002 | ns | ns | 0.00 | –0.001 |
| Constant | **–2.730** | –0.840 | –0.780 | **–1.97** | –1.010 |
| $R^2$ | 0.387 | 0.453 | 0.453 | 0.407 | 0.392 |

*Notes*: Variables significant at $p < 0.01$ in **bold**; variables significant at $p < 0.05$ or $p < 0.1$ in regular type; ns = not significant at 0.01 level, suppressed. Distances, unless otherwise noted, in tenths of a mile. See Table 3 for notes on variables included but not shown.

mile away is associated with lower appreciation for those living in higher-income tracts, which is a counter-intuitive result because higher-income families are more likely to own and use autos. Whistleblowing points and distance to track are associated with property value increases prior to operation and with roughly offsetting decreases after operation; while puzzling individually, the net effects are neutral.

Next, we tested whether the line's effects seem to be different for smaller and larger houses (Table 5, columns 3–5). We measured size with the number of bedrooms, because square footage and lot size are inconsistently available in the dataset. About 20 per cent of houses have one to two bedrooms; 51 per cent have three bedrooms; and 29 per cent have four or more.

Within a quarter-mile of stations, small homes have a net appreciation rate 42 per cent higher than comparable properties (50—7.9 per cent), but this difference along with others in this sub-sample is not statistically significant, as there are just 18 houses with sales post-operation in this radius (Table 5, column 3).

The *net* effect is smaller or non-existent for high-ridership stations. In the quarter-mile to half-mile ring there is another insignificant but net 7 per cent positive estimate, while for distances from a half-mile to five miles away the differences are smaller and insignificant, but negative on net.

For three-bedroom houses (Table 5, model 4), with a significantly larger sample size of 87 homes within a quarter-mile, we find similar but slightly smaller effects. For three-bedroom houses with stations from a quarter-mile to half-mile away, there is a net positive 23 per cent appreciation (41-15 per cent), although only at the 95 per cent confidence level. This is dampened for properties near busier stations so that for most properties the net effect is neutral or negative. The net effect for three-bedroom homes is a positive but statistically insignificant 10 per cent out to a half-mile away. Beyond that distance the effect is slightly negative, but not statistically significant. Although the statistical significance is low, the correlation pattern could reflect a market premium for smaller houses near stations that reduces the bidding market for houses farther away from stations.

Finally, houses of four or more bedrooms right near the River Line apparently experience no effect of the line's groundbreaking or operation. However, those within one-quarter and 2 miles have statistically significant net depreciation (Table 5, model 5). For example, a four-bedroom house one mile away from a station that sold before groundbreaking and after operation has net 10 per cent depreciation compared with a home more than five miles away, accounting for the gradient and the threshold effects. The sums of coefficients farther out are not statistically distinguishable from zero.

### 4.3 Comments on the Analysis

We considered carrying out spatially weighted regression or other non-parametric techniques to control for spatial autocorrelation (for example, Redfearn, 2009; McMillen, 2010). However, these methods are most relevant for cross-sectional data with relatively few independent variables and the suspicion of many unobserved spatially distributed variables (for example, Chalermpong, 2007). The techniques do not always perform better than ordinary least squares (Martínez and Viegas, 2009). Repeat sales data provide better control over unobserved heterogeneity and, in addition, we employ many spatial and other independent variables. If omitted variable bias causes spatial dependence among the errors, spatial statistics are a poor substitute to including the omitted variables (Tse, 2002).

This analysis does not address possible issues of blight and abandonment in the study area, although the results are robust to such issues. Camden lost 157 000 jobs from 1950 to 2001 and the city's population declined from a peak of about 125 000 to about 80 000 presently (Lake *et al.*, 2007). Our estimates are only on homes transacted via the Multiple Listing Service and therefore are not likely to include many properties that are deteriorated and declining in value. We also included municipal and county fixed effects variables, distance to nearest central business district (including Camden and Trenton), income, percentage African American, residential vacancy and school quality variables that are likely to help to control for urban structure, blight, abandonment and other potential causes of property value changes; and we drew properties from a very large area, including parts of all of the municipalities that are not near rail stations.

We tested for only groundbreaking and operation effects on property appreciation and not any possible effects of the initial announcement of the line. We believe it unlikely that larger gains in property prices could have occurred after the plans for the line became public but prior to groundbreaking, for two reasons. First, there was a great deal of uncertainty about the siting of the

line up until groundbreaking and there was some substantial negative press, as described earlier. Secondly, tentative plans for a light rail line could have been reflected in land prices but are much less likely to be reflected in the willingness to bid on a house in the consumer-side real estate market.

## 5. Conclusions

> Perhaps the real significance of transport is that it restructures the economy, which may or may not lead to enhanced growth (Vickerman, 2001, p. 636).

Our results suggest that short-term economic impacts in the owned housing market do little to help justify the River Line investment. The cumulative net effect across all owned housing units in the five-mile radius around stations is slightly negative or at best neutral. However, owners of small homes and homes in lower-income census tracts may have benefited. If apartment renters near stations are similar to owners of small homes, they may also have benefited. Firms near stations could also experience substantially more positive impacts than homeowners, depending on their labour force and customers. Crucially, however, these limited benefits to sub-markets near stations might be realised through a sorting process that also causes a reduction in property values for properties somewhat farther away from stations. The economic impacts of the River Line may be primarily redistributive. Net economic impacts could be increasingly positive over time if more and higher-density housing could be developed nearby.

This is also one of the few studies to distinguish announcement and operation effects (see also Du and Mulley, 2007). The River Line may have had positive effects on property appreciation after it started operating, but that these effects did not generally make up for property depreciation that occurred after groundbreaking and prior to operation. The negative groundbreaking effects could be partly due to construction impacts and partly due to buyer anticipation that the River Line would cause crime and traffic to rise near stations. Operation effects appear to have been mostly positive, as properties regained some of the lost value after the line began operating, suggesting these fears were proven partly untrue once the line started to operate. This is in striking contrast to the McMillen and McDonald (2004) study, which found positive announcement effects and attributed their lower magnitude to discounting and uncertainty. Media coverage and expectations about the line may have contributed to depreciation effects and perhaps current net valuations do not yet fully reflect the positive accessibility value of the Line. More explicitly than past studies of this kind, our study suggests an important influence upon the land market of public and media perceptions of the planning, construction and implementation process.

This study has some methodological implications for other efforts using property values to study the economic impacts of rail. First, focusing on subsets of the market gives strikingly different results. Even low-ridership rail systems, such as the River Line, could have significant positive impacts on properties of the right type. Indeed, renters and commercial property owners may be the primary beneficiaries of rail investments, but most studies (like this one) are limited to owned homes. Also, apparent regional differences in capitalisation rates for rail in cross-sectional studies might be only partly due to differences in transit accessibility characteristics. A different housing stock mix, implying different households with varying values for access to transit, might also play a role. Secondly, the repeat sales method may explain some variance with other cross-sectional studies. For example, our findings are at odds with several studies finding that rail stations in lower-income Census tracts have smaller positive impacts

or even negative impacts. Such studies may suffer from omitted variable bias. Thirdly, the tentative evidence of transfers of appreciation value from farther-away properties to close-in properties suggests that even among smaller houses or condominiums and houses in lower-income neighbourhoods, studying only properties within a mile or two of rail might overstate the (net) positive impacts.

The results also have implications for planning. On the one hand, they reinforce the notion that, when choosing alignments, rail planners should evaluate nearby housing stock and neighbourhoods, availability of developable parcels and zoning constraints. Smaller houses and lower-income areas are more likely to benefit and changes over time to permit more lower-cost housing could intensify the benefit. In doing so, planners must also of course consider the potential for displacement via increasing rents, as noted earlier; areas with low ownership levels are naturally more susceptible. On the other hand, the fact that larger homes do not appear to have benefited is problematic. Higher-income households are politically influential and must in many cases be satisfied to achieve political support for rail investments.

## Notes

1. The dummy variables representing distance ranges (for example, one-quarter-mile, one-half-mile, 1 mile, etc.) to stations, track and whistle-blowing points ranged in value from 0.003 to 0.05. They have been omitted from Table 2 for clarity.
2. These are collinear by construction but do not present estimation problems as statistical significance remains high given the large sample size.
3. We conducted a variance inflation test to confirm that the change in sign is not due to multicollinearity.
4. Because we are not attempting to distinguish flood risk from water-related amenities (for example, Daniel *et al.*, 2009), but rather to control for any other effects that might be correlated with station access or track distance, this specification is sufficient for our purposes.

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